Output Ontology (OO)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: *CRISISBENCH* | ***Context:*** *Argentina — Buenos Aires Policy Maker Cohort (Disaster Reporting Reliability)*

Output Ontology definition: Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Output ontology is HIGH priority per elicitation. The two implemented tasks — binary informativeness and multi-class humanitarian type [Q12] — provide a usable foundation but do not natively include the source credibility, misinformation likelihood, or signal-to-noise reliability dimension that is the deployment's core requirement [elicitation Q3]. The user explicitly acknowledges this gap and intends to construct credibility scoring downstream by combining model outputs with external metadata, but the benchmark's data schema (id, event, source, text, lang, lang_conf, class_label) contains no tweet-level credibility metadata to learn from (CRITICAL Concern 2). Dataset analysis empirically confirms the disconnect: a 'DEATH TOLL Now under Investigation for COVER UP!!!' tweet labelled not_informative [DATASET-D45] and a Boston Bombing conspiracy tweet labelled informative [DATASET-D47] illustrate that informativeness ≠ credibility. The label space is also imbalanced even after consolidation (37.40% 'not humanitarian' [Q56]), with low-support classes [Q52, Q53, Q54] and class-mapping discrepancies that produce results reported only for classes the model can classify [Q88]. TREC-IS provides the closest precedent (Low/Medium/High/Critical priority labels) but covers no Argentine events [WEB-15].

ID	Evidence
Q3	“Typical classification tasks in the community include (i) informativeness (i.e., informative vs. not-informative messages), (ii) humanitarian information types (e.g., affected individual reports, infrastructure damage reports), and (iii) event types (e.g., flood, earthquake, fire).” (p.1)
Q12	“We provide benchmark results on English tweets set using state-of-the-art machine learning algorithms such as Convolutional Neural Networks (CNN), fastText (Joulin et al. 2017) and pre-trained transformer models (Devlin et al. 2019) for two classifications tasks, i.e., Informativeness (binary) and Humanitarian type (multi-class) classification.” (p.2)
Q52	“Only 17 tweets with the label "terrrorism related" are present in CrisisNLP.” (p.5)
Q53	“Similarly, the class "disease related" only appears in CrisisNLP.” (p.5)
Q54	“The scarcity of the class labels poses a great challenge to design the classification model using individual datasets.” (p.5)
Q56	“For example, the distribution of "not humanitarian" is relatively higher (37.40%) than other class labels.” (p.5)
Q88	“In the humanitarian task, for different datasets in Table 6, we have different number of class labels. We report the results of those classes only for which the model is able to classify.” (p.8)
D45	https://t.co/najVJNt4Ip DEATH TOLL Now under Investigation for COVER UP!!! https://t.co/VuGGPYboIv — Credibility-ambiguous tweet about death toll cover-up labelled not_informative
D47	I was told I'm ignorant to think the Boston Bombing wasn't planned by the government to keep our attention away from schemes being planned. — Conspiracy tweet labelled informative
WEB-15	TREC-IS introduced priority/criticality labels (Low/Medium/High/Critical) as the closest existing benchmark precedent for credibility scoring; covers no Argentine events (https://trecis.github.io/)

Strengths

- Carefully documented label-mapping process across 8 source datasets [Q33, Q106–Q110, Q123–Q125] enables transparent inspection of the consolidated taxonomy
- Top-level humanitarian categories are oriented toward humanitarian aid relevance and align with internationally-normed disaster taxonomies Argentine professionals reference [Q18]

- Binary informativeness layer functions as a deployable first-pass triage filter — operationalizes Step 1 of the user's downstream credibility pipeline [DATASET-D34]

ID	Evidence
Q18	“In this study, we use the class labels that are important for humanitarian aid for disaster response tasks, which are common across the publicly available resources.” (p.2)
Q33	“The datasets come with different class labels. We create a set of common class labels by manually mapping semantically similar labels into one cluster. For example, the label "building damaged," originally used in the AIDR system, is mapped to "infrastructure and utilities damage" in our final dataset.” (p.3)
Q106	“In Table 12, we report class label mapping for ISCRAM2013, SWDM2013, CrisisLex and CrisisNLP datasets.” (p.11)
Q107	“Note that all humanitarian class labels also mapped to informative, and not humanitarian labels mapped to not-informative in the data preparation step.” (p.11)
Q108	“In Table 13, we report the class label mapping for informativeness and humanitarian tasks for DRD dataset.” (p.11)
Q109	“The DSM dataset only contains tweets labeled as relevant vs not-relevant, which we mapped for informativeness task as shown in Table 14.” (p.11)
Q110	“The CrisisMMD dataset has been annotated for informativeness and humanitarian task, therefore, very minor label mapping was needed as shown in Table in 15.” (p.11)
Q123	“Table 12: Class label mapping and grouping for CrisisLex, CrisisNLP, ISCRAM2013, and SWDM2013 datasets. The symbol (✗) indicates we do not map the tweets with that label for this study.” (p.13)
Q124	“The symbol (✗) indicates we do not map the tweets with that label for this study.” (p.14)
Q125	“Table 16: Class label mapping for AIDR system.” (p.15)
D34	RT @cityofcalgary: Acting Fire Chief, Ken Uzeloc, will address the media at Ogden Road and 17 Street S.E. at 10 p.m. #yyc #yycflood — Official municipal announcement — clear informative label, Canadian context

Checklist

Checklist item definitions:

ID	Assessment Question
OO-1	Review output label categories for regional relevance
OO-2	Identify missing regionally specific categories
OO-3	Flag categories encoding non-regional values
OO-4	Consider stakeholder-driven taxonomy redesign
OO-5	Document taxonomy issues harming structural/content validity

Checklist responses:

OO-1: Top-level humanitarian categories (affected_individual, infrastructure_damage, requests_or_needs, response_efforts, caution_and_advice, displaced_and_evacuations, donation_and_volunteering) are regionally relevant for Argentine emergency triage. The informativeness binary is also relevant.

OO-2: Critical missing category: source credibility / signal-reliability dimension absent from output space [DATASET CRITICAL Concern 2]. Argentine hazard sub-labels (pampero, sudestada) and villa-impact framing are absent. TREC-IS priority labels (Low/Medium/High/Critical) are the closest existing precedent but not part of CrisisBench [WEB-15].

OO-3: Some categories encode non-regional framing: 'terrorism related' (17 instances [Q52]) and 'disease related' (CrisisNLP only [Q53]) reflect specific source-event distributions. The 'not humanitarian' class at 37.40% [Q56] dominates and may inherit North American assumptions about what counts as humanitarian-relevant.

OO-4: Stakeholder-driven taxonomy redesign is warranted given the user-acknowledged credibility-scoring gap. The user's downstream construction approach (combining model outputs with external metadata) is a partial mitigation but not a benchmark-level fix.

OO-5: Documented taxonomy issues: (a) absent credibility/reliability dimension (structural validity violation for the deployment's core construct); (b) absent Argentine sub-labels; (c) heterogeneous 'other_relevant_information' catch-all (Concern 7); (d) imbalanced classes with very low minority support [Q52–Q56].

ID	Evidence
Q12	“We provide benchmark results on English tweets set using state-of-the-art machine learning algorithms such as Convolutional Neural Networks (CNN), fastText (Joulin et al. 2017) and pre-trained transformer models (Devlin et al. 2019) for two classifications tasks, i.e., Informativeness (binary) and Humanitarian type (multi-class) classification.” (p.2)
Q18	“In this study, we use the class labels that are important for humanitarian aid for disaster response tasks, which are common across the publicly available resources.” (p.2)
Q52	“Only 17 tweets with the label "terrrorism related" are present in CrisisNLP.” (p.5)
Q53	“Similarly, the class "disease related" only appears in CrisisNLP.” (p.5)
Q54	“The scarcity of the class labels poses a great challenge to design the classification model using individual datasets.” (p.5)
Q55	“Even after combining the datasets, the imbalance in class distribution seems to persist (last column in Table 4).” (p.5)
Q56	“For example, the distribution of "not humanitarian" is relatively higher (37.40%) than other class labels.” (p.5)
Q88	“In the humanitarian task, for different datasets in Table 6, we have different number of class labels. We report the results of those classes only for which the model is able to classify.” (p.8)
WEB-15	TREC-IS introduced priority/criticality labels (Low/Medium/High/Critical) as the closest existing benchmark precedent for credibility scoring; covers no Argentine events (https://trecis.github.io/)
WEB-21	2024 NLP credibility survey: aggregating multiple credibility signals into a unified score remains fragmented and underspecified (https://arxiv.org/html/2410.21360v2)

Information Gaps

- Whether downstream credibility-scoring pipelines built on top of this benchmark's outputs achieve operationally useful F1 in Argentine deployment
- Whether external metadata (institutional affiliation, source popularity) the user envisions combining with model outputs is reliably available in Argentine post-Télam press environment

Requires Expert Verification

- Argentine emergency management expert validation of which output categories are operationally critical vs. nice-to-have
- Whether TREC-IS priority labels would be a useful supplementation target for Argentine deployment

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Remediation

Gap 1: No source credibility / signal-reliability dimension in the output space, which is the deployment's core requirement

Recommendation: Construct a downstream credibility-scoring layer combining CrisisBench classifier outputs with Argentine source-metadata streams (post-Télam press source registry, institutional affiliation, AFE/SINAGIR official channels, account history). Use TREC-IS priority labels (Low/Medium/High/Critical) [WEB-15] as the target output schema; align with Buenos Aires Province Resolución 4/2025 [WEB-11] for regulatory compliance.

ID	Evidence
WEB-11	https://regulations.ai/regulations/argentina-summary
WEB-15	TREC-IS introduced priority/criticality labels (Low/Medium/High/Critical) as the closest existing benchmark precedent for credibility scoring; covers no Argentine events (https://trecis.github.io/)